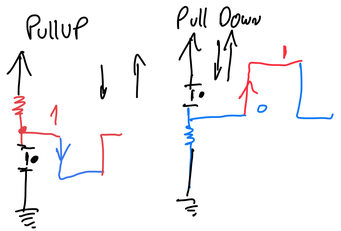
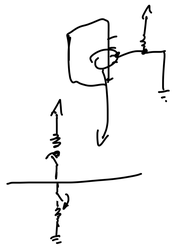
Entradas Digitales →

- RTOS



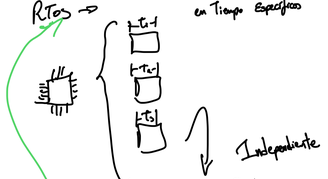


Super Bucle

```
int main{
while(0){
}
return 0;
}
```

```
While {
Leer Sensor
Conexion HTTP
Leer Botones
}
```

RTOS →



Ejecuta Tareas en Tiempo Especifico

Independiente

Tarea = Task

- Tarea Sensor (10ms)
- Tarea ConSock
- Tarea VI Lugl
- Tarea Send HTTP

Manejo → RTOS → Kernel ligero

- ↳ Scheduler → Planificador de Tareas
- ↳ Task / Threads
- ↳ Semaforos (Binario / Conteo) / Mutex
- ↳ Timers / Manejo ISR
- ↳ Queue

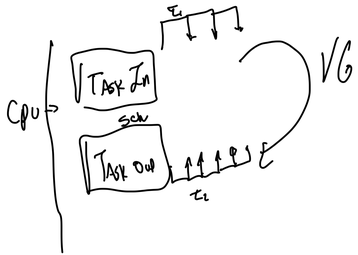
Realtime-ish
- Hard → X Fallar el Tiempo
- Soft → Tolerar Pequeños Retrasos

Esp-IDF → FreeRTOS → RTOS

- AMD → Microblaze
- Intel → Nios II
- Renesas
- STM32
- Texas Inst
- NXP
- AVR

Zephyr → Linux Foundation

ARM

BLE - Wifi - LoRa - Thread

LoRaWAN - Matter